



KEMET

AI-Powered Egypt Tourism Assistant

An intelligent tourism ecosystem that combines AI, personalized trip planning, real-time travel insights, and curated cultural experiences for exploring Egypt

Team Members



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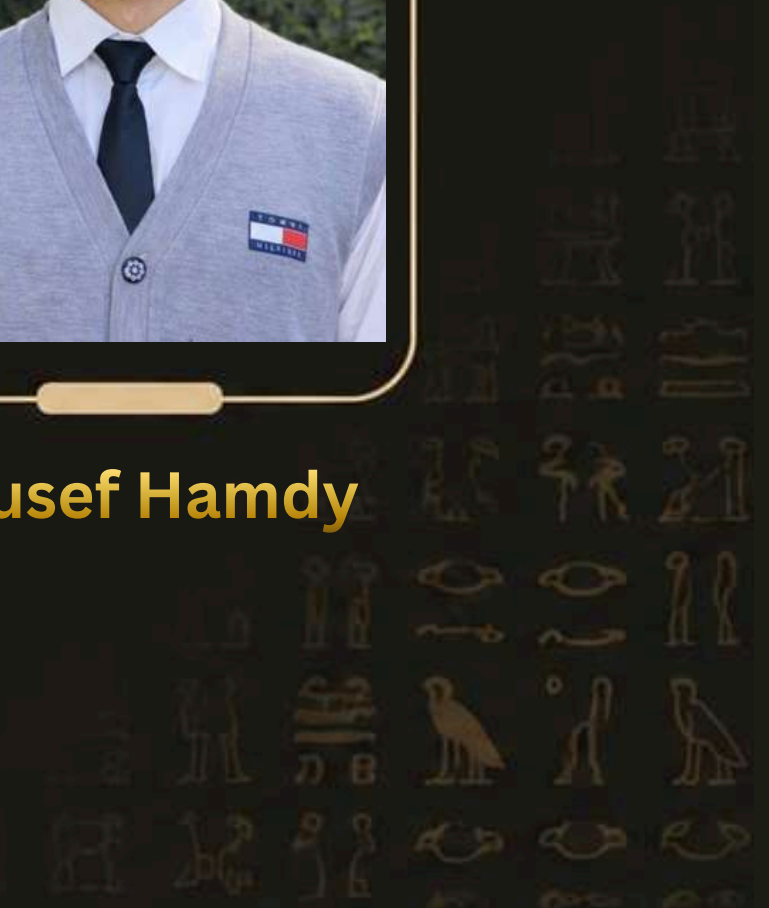
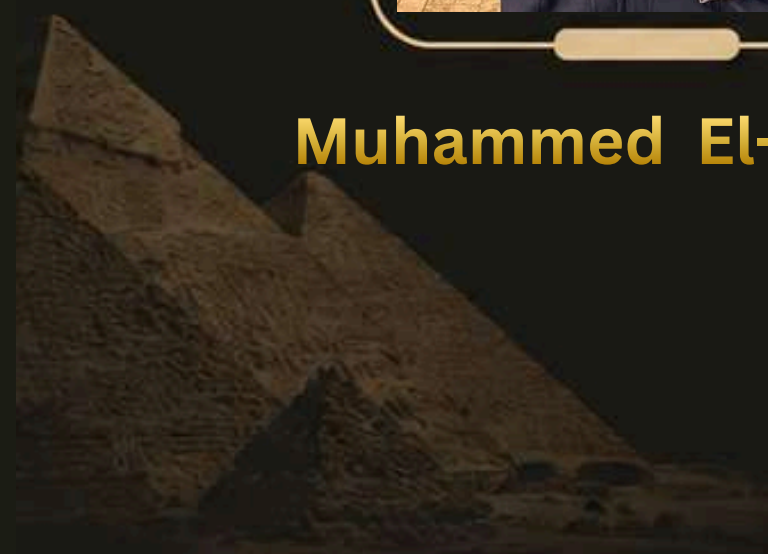
Elsayed Ashraf



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OVERVIEW





What is KEMET?

AI-Powered Egypt Tourism Assistant

KEMET is an intelligent tourism platform that helps travelers discover Egypt through one unified application. It combines Artificial Intelligence, live tourism information, curated datasets, and personalized trip planning to provide a seamless travel experience.



AI Chatbot

Get smart, accurate answers about Egypt in seconds.



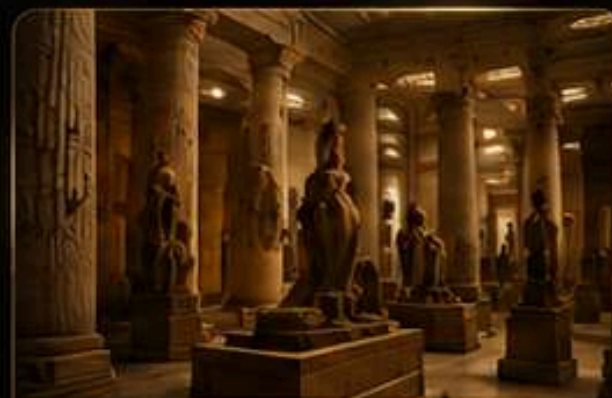
Smart Trip Planner

Plan your perfect trip with personalized recommendations.



Live Dashboard

Real-time weather, arrivals, exchange rates & more.



Tourism Directories

Explore hotels, restaurants, museums, monuments & more.



Community Platform

Connect, share experiences and discover from travelers.



Location Explorer

Discover places on the map and explore by location.



12+

Interactive Modules



1200+

Tourism Records



AI + RAG

Powered



Real-Time

Live Data





PROBLEM STATEMENT





Why KEMET?

What prevents tourists from experiencing the best of Egypt?



Fragmented Information

Tourism information is spread across multiple websites and platforms.

Difficult Trip Planning

Travelers manually organize hotels, attractions and transportation.

Limited Trusted Guidance

Finding verified guides and authentic recommendations is challenging.

Lack of Real-Time Updates

Tourists struggle to access live travel information and timely recommendations.



HOW TO USE?





FROM PLANNING TO EXPLORATION

Supporting every stage of a tourist's journey in Egypt

01

BEFORE THE TRIP

Discover Egypt and get instant travel guidance.

AI Chatbot

Tourism Dashboard



02

PLAN YOUR JOURNEY

Create a personalized itinerary tailored to your interests, budget, and travel style.

AI Trip Planner

Hotels

Restaurants



03

EXPLORE EGYPT

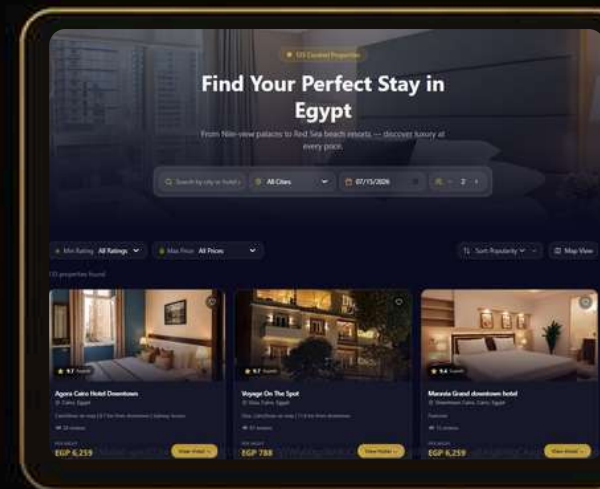
Discover museums, monuments, archaeological sites, and historical landmarks through verified tourism data.

Museums

Monuments

Historical Sites

Historical Periods



04

TRAVEL WITH CONFIDENCE

Access trusted local services, live information, and cultural tools throughout your journey.

Certified Guides

Muslim Assistant

Maps & Navigation



05

SHARE YOUR EXPERIENCE

Connect with the community by sharing travel stories, photos, and recommendations.

Community

User Posts & Reviews





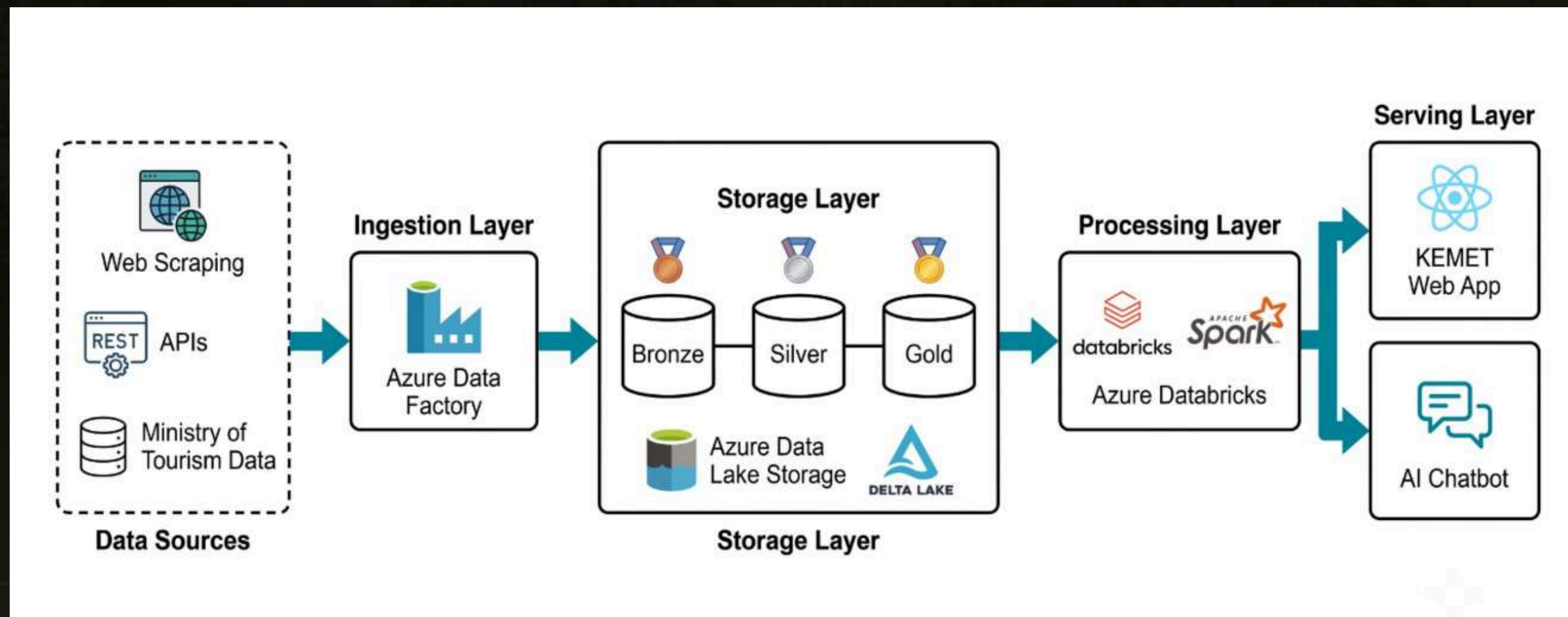
TECHNOLOGY AND PIPELINE





Data Engineering Pipeline

The KEMET platform is powered by a cloud-native data engineering pipeline that collects tourism data from multiple trusted sources, processes it using Azure Data Factory and Azure Databricks, and organizes it through the Medallion Architecture before serving clean, analytics-ready data to our Flask APIs and AI-powered web platform





Data Collection & Ingestion

Raw Data was gathered from trusted sources using web scraping (Selenium), and public APIs, creating a reliable and continuously updated knowledge base for KEMET.

- Hotels & Accommodation Data
- Restaurants & Local Dining
- Museums
- Historical Landmarks
- Monuments
- Archaeological Sites



Booking.com



Google Maps



Tripadvisor



AZURE CLOUD ARCHITECTURE

Scalable Cloud Platform for Tourism Data Engineering



DATA SOURCES

-  Web Scraping
-  Google Maps
-  Booking.com
-  Tripadvisor
-  Ministry of Tourism



AZURE DATA FACTORY



- Pipeline orchestration
- Automated ingestion
- Data scheduling



AZURE DATA LAKE STORAGE GEN2



- Centralized storage
- Secure and scalable
- Bronze, Silver, Gold layers



AZURE DATABRICKS

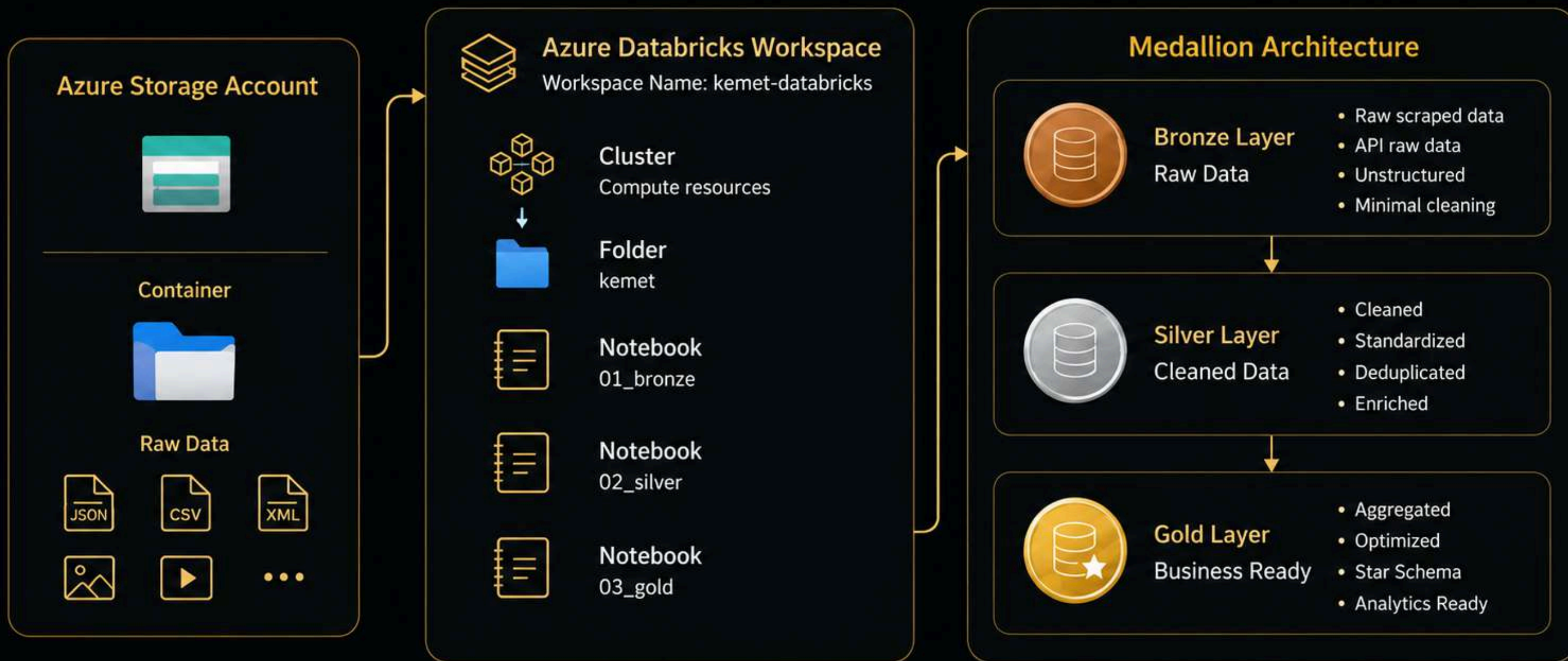


- Distributed data processing
- PySpark transformations
- Delta Lake for reliability



Data Processing & Medallion Architecture

Transforming raw tourism data into analytics-ready insights

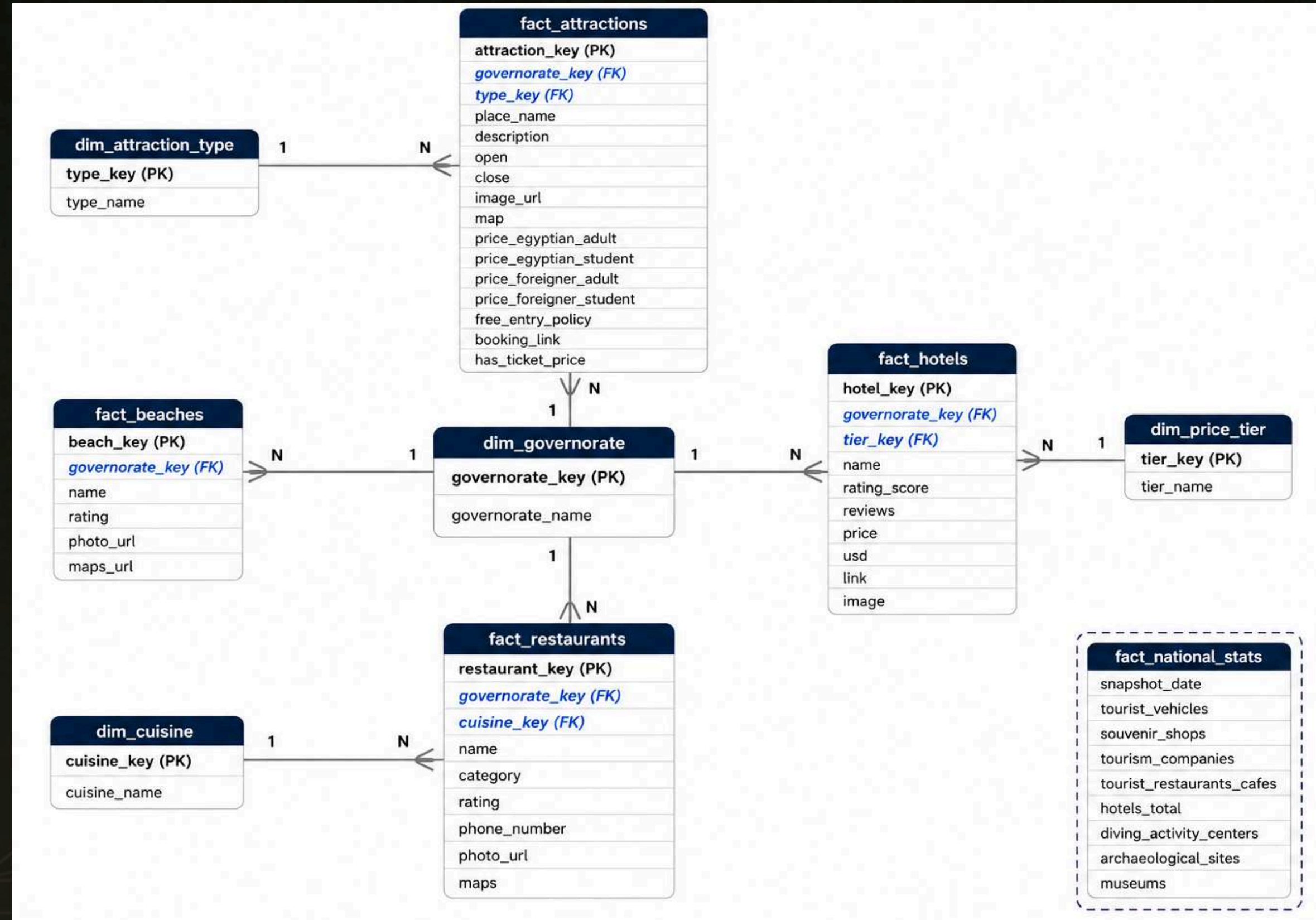




Galaxy Schema

Key Benefits :

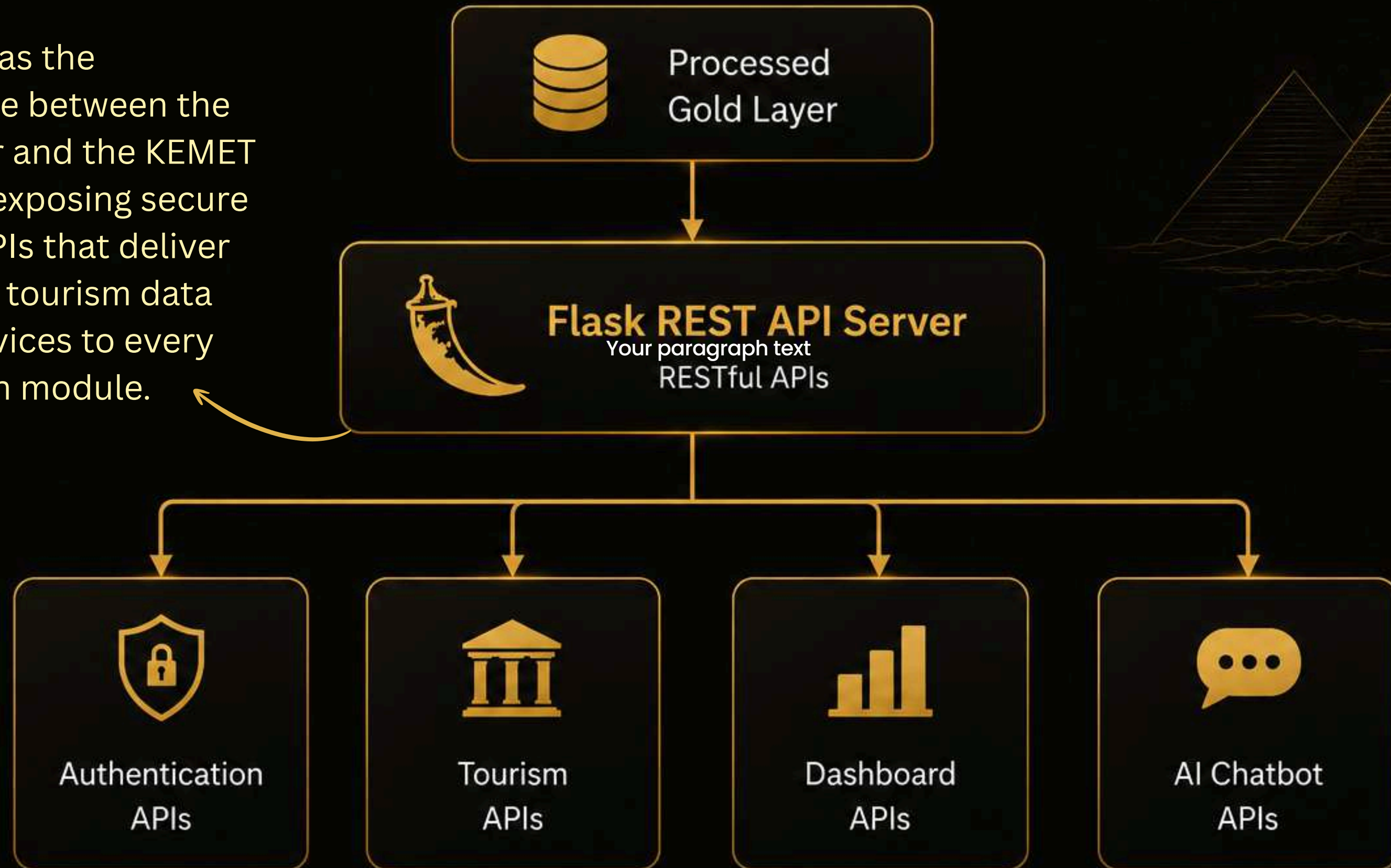
1. Fast analytical queries
2. Centralized tourism data
3. Simplified reporting
4. Scalable data model
5. Supports AI & dashboards





Backend API Layer

- **Flask** acts as the middleware between the Gold Layer and the KEMET platform, exposing secure RESTful APIs that deliver processed tourism data and AI services to every application module.





introducing..

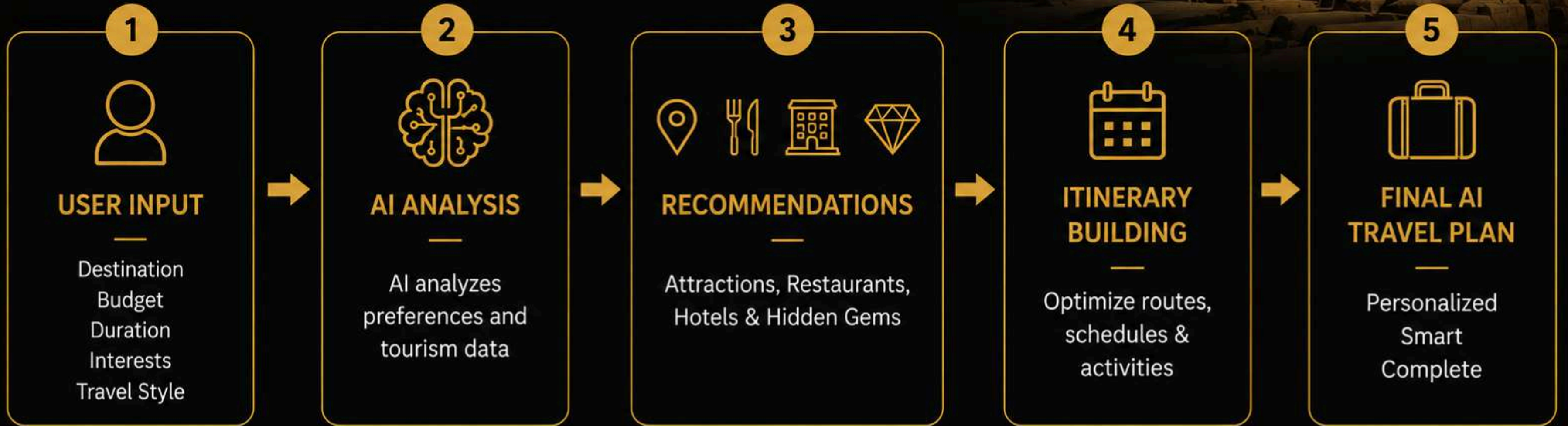
KEMET AI





AI RECOMMENDATION ENGINE

Turning Tourism Data into Personalized Experiences





RETRIEVAL-AUGMENTED GENERATION (RAG)

Using our tourism knowledge base to deliver accurate, context-aware answers.



KNOWLEDGE SOURCES



HOTELS



RESTAURANTS



MUSEUMS



MONUMENTS



ATTRACTIONS



MINISTRY OF TOURISM



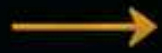
AI CHATBOT ARCHITECTURE

Natural Language Interaction Powered by Gemini AI



TOURIST QUESTION

User asks a question
in natural language



FLASK API

Receives and
validates the
user request



PROMPT PROCESSING

Prepares and
structures the
prompt



GEMINI 2.0 FLASH MODEL

Generates intelligent
response using AI
capabilities



AI RESPONSE

Context-aware,
relevant and
personalized answer



RESPONSE TO USER

Answer delivered
back to the user
through the API



WHAT TO EXPECT IN THE FUTURE





FUTURE IMPROVEMENTS



Continuously evolving KEMET to build a smarter and more intelligent tourism experience.



01. Mobile Application

Develop native Android and iOS apps for a seamless travel experience.



02. Real-Time Data Streaming

Integrate live events, traffic, weather and tourism updates using Azure Event Hubs and Stream Analytics.



03. Smarter AI Recommendations

Enhance personalization by learning from user preferences, travel history and behavioral patterns.



04. Interactive Maps & Navigation

Provide real-time navigation, route optimization, nearby attractions and location-aware recommendations.



05. Scalable Cloud Data Pipeline

Expand the Azure architecture with automated orchestration, monitoring and incremental data processing.



06. Predictive Tourism Analytics

Apply machine learning to forecast tourist demand, seasonal trends and attraction popularity.





THANK YOU!

ANY QUESTIONS?

